Higher order derivatives

We can take partial derivatives of partial derivatives and these we call 2^{nd} order partial derivatives

e.g. $f(x,y) = x^2 y + e^{x+y}$ compute 2^{nd} order partial derivatives

$$\frac{\partial f}{\partial x}(x,y) = 2xy + e^{x+y}$$
$$\frac{\partial f}{\partial y}(x,y) = x^2 + e^{x+y}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)(x,y) = 2y + e^{x+y}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)(x,y) = e^{x+y}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)(x,y) = 2x + e^{x+y}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)(x,y) = 2x + e^{x+y}$$

these we call crossed or mixed 2^{nd} order partial derivatives

Thm: If f in C^2 (all 2^{nd} order part. der. are continuous) then $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$.

e.g. Verify that $f(x,y,z) = 2x^2 - y^2 - z^2$ satisfies $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$.

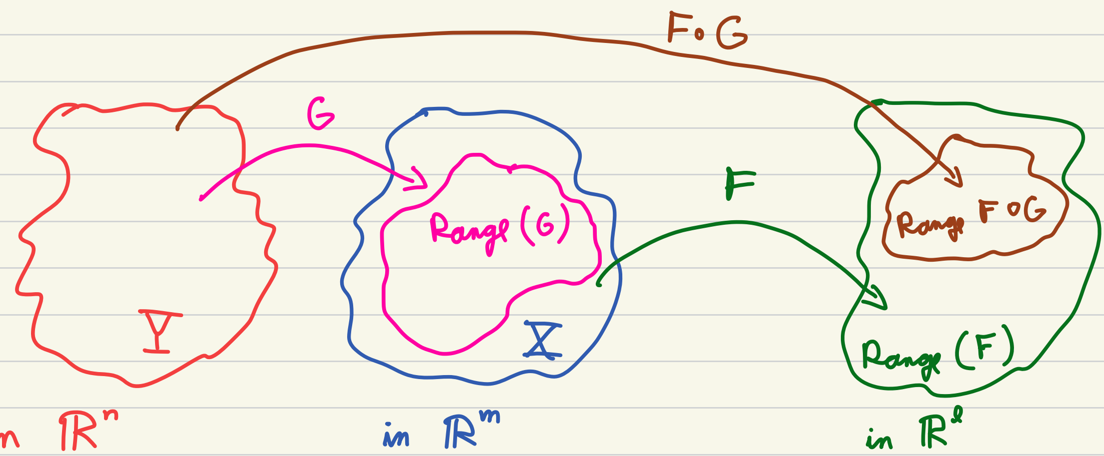
The Chain Rule

February 23, 2026

Recall from Calc I: If $g: Y \subseteq \mathbb{R} \to \mathbb{R}$ and $f: X \subseteq \mathbb{R} \to \mathbb{R}$
both f and g differentiable and such that $\text{Range}(g) \subseteq X$,
then $f \circ g : Y \subseteq \mathbb{R} \to \mathbb{R}$ is differentiable and
$(f \circ g)'(t) = f'(g(t)) \cdot g'(t)$

$\Bigg($ we can also think of it as $\begin{array}{l} g: Y \to \mathbb{R} \\ t \mapsto g(t) = x(t) \end{array}$ $\quad$ $\begin{array}{l} f: X \to \mathbb{R} \\ x \mapsto (x) \end{array}$

$$\frac{d}{dt}(f \circ g)(t) = \frac{df}{dx}\bigg|_{g(t)} \cdot \frac{dg}{dt}(t) = \frac{df}{dx}\frac{dx}{dt}$$
$\Bigg)$



Chain rule: Suppose $G: Y \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$
and $F: X \subseteq \mathbb{R}^m \longrightarrow \mathbb{R}^l$
with $Y \subseteq \mathbb{R}^n$ open, $X \subseteq \mathbb{R}^m$ open and $\text{Range}(G) \subseteq X$.
If G is differentiable at $\vec{a}$ and F is differentiable at $G(\vec{a})$,
then $F \circ G$ is differentiable at $\vec{a}$ and

$$D(F \circ G)\big|_{\vec{a}} = DF\big|_{G(\vec{a})} \cdot DG\big|_{\vec{a}}$$

matrix multiplication

e.g.: $g(x) = e^{2x}$ and $f(t) = t^2 - 1 = x(t)$ $\qquad g \circ f$

pictorially

$$
\begin{array}{ccccc}
\mathbb{R} & \xrightarrow{f} & \mathbb{R} & \xrightarrow{g} & \mathbb{R} \\
t & \longmapsto & x = f(t) & \longmapsto & g \circ f(t)
\end{array}
$$

final independent variable $\qquad$ intermediate variable $\qquad$ dependent variable

$$\frac{d}{dt}(g(f(t))) = \frac{dg}{dx}\,\frac{dx}{dt}$$

$$\frac{dg}{dx} = 2e^{2x} \qquad \frac{df}{dt} = \frac{dx}{dt} = 2t$$

$$\frac{d}{dt}(g(f(t))) = 2e^{2(f(t))} \cdot 2t = 2e^{2t^2-2} \cdot 2t$$

$$= 4te^{2t^2-2}$$

eg: $f(x,y) = e^{2x+y}$ $\qquad x = t+1 \qquad y = 3t-4$

$$\frac{df}{dt} = D\big(F(x(t), y(t))\big)$$

$$
\begin{array}{ccccc}
\mathbb{R} & \xrightarrow{(x,y)} & \mathbb{R}^2 & \xrightarrow{f} & \mathbb{R} \\
t & \longmapsto & (x(t), y(t)) & \longmapsto & f(x,y)
\end{array}
$$

indep. var. $\qquad$ intermediate variable $\qquad$ dependent var.

long path $\quad f(x(t), y(t)) = e^{2(t+1)+3t-4} = e^{5t-2} \qquad \frac{df}{dt} = 5e^{5t-2}$

chain rule : $Df\big|_{(x(t),y(t))} \cdot D(x(t), y(t))\big|_t$

$$Df = \vec{\nabla} f = (2e^{2x+y}, e^{2x-y}) \qquad D(x,y)\big|_t = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

matrix $\qquad$ vector

$$= \left(2e^{2(t-1)+(3t-4)}, e^{2(t+1)+(3t-4)}\right) \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 5e^{5t-2}$$

eg: $F(x,y) = \begin{pmatrix} x \\ x^2+y \\ 7 \\ \cos(y) \end{pmatrix}$

$G(u,v,w) = \begin{pmatrix} u \\ 2v - w^2 \end{pmatrix}$

recall $DF \in M_{4\times 2}$
$DG \in M_{2\times 3}$

$$D(F\circ G) = DF\big|_{G(u,v,w)} \cdot DG\big|_{(u,v,w)}$$

$$DF\big|_{(x,y)} = \begin{pmatrix} 1 & 0 \\ 2x & 1 \\ 0 & 0 \\ 0 & -\sin(y) \end{pmatrix}$$

$$DG\big|_{(u,v,w)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2w \end{pmatrix}$$

$$D(F\circ G)\big|_{(u,v,w)} = \begin{pmatrix} 1 & 0 \\ 2x & 1 \\ 0 & 0 \\ 0 & -\sin(y) \end{pmatrix}\Bigg|_{G(u,v,w)} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2w \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 2u & 1 \\ 0 & 0 \\ 0 & -\sin(2v-w^2) \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -2w \end{pmatrix}$$

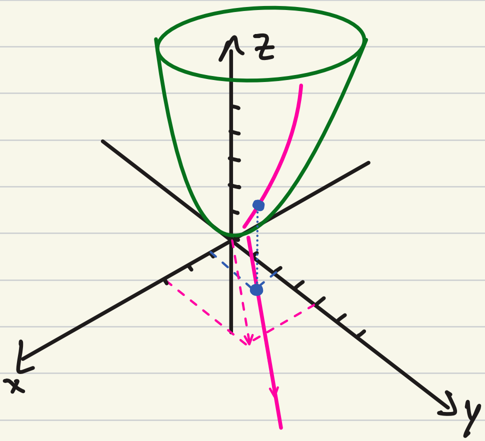
$$= \begin{pmatrix} 1 & 0 & 0 \\ 2x & 2 & -2w \\ 0 & 0 & 0 \\ 0 & -2\sin(2v-w^2) & 2w\sin(2v-w^2) \end{pmatrix}$$

"aside we have"
$x = u$
$y = 2v - w^2$

$F\circ G(u,v,w)$
$= F(x(u,v,w), y(u,v,w))$

e.g. $f(x,y) = x^2 + y^2$

$$x(t) = 3t + 1$$
$$y(t) = 4t + 2$$



$D(f(x(t), y(t)))$

$Df = \begin{pmatrix} 2x & 2y \end{pmatrix}$

$D(x,y) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

evaluate at $t=0$

$$D(f \circ (x,y)(t)) = \begin{pmatrix} 2(1) & 2(2) \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$= 6 + 16 = 22$$

$\frac{df}{dt} = 22$

speed of $(x(t), y(t))$ motion at $t=0$
is $s = \sqrt{3^2 + 4^2}$

normalizing $\frac{df}{dt}$ by the speed we get $\frac{22}{5}$

$$\vec{\nabla f}\Big|_{(1,2)} \cdot \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} = \frac{22}{5}$$

derivative of f as we pass the point $(1,2)$
with velocity in the direction $(3,4)$ and speed 1 is $\frac{22}{5}$